Divide and Conquer

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The **divide-and-conquer** paradigm,

Divide the problem into one or more subproblems that are smaller instances of the same problem.

Conquer the subproblems by solving them recursively.

Base case: If the subproblems are small enough, just solve them by brute force.

Combine the subproblem solutions to form a solution to the original problem.

Recursion Tree

METHODS FOR SOLVING RECURRENCES

- **Substitution method:** Guess the solution, then use induction to prove that it's correct.
- **Recursion-tree method:** Draw out a recursion tree, determine the costs at each level, and sum them up. Useful for coming up with a guess for the substitution method.

Recursive-tree method

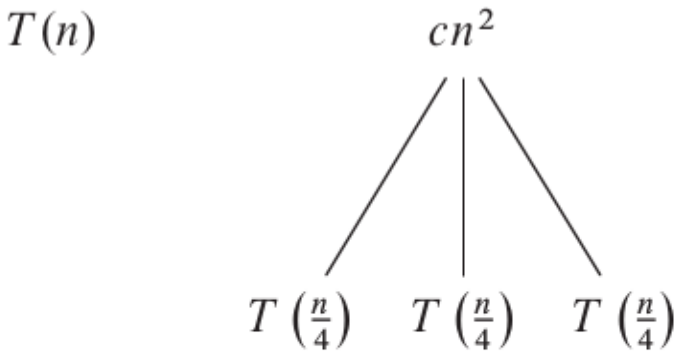
- Draw out a recursion tree, determine the costs at each level, and sum them up.
- Useful for coming up with a guess for the substitution method.
- If you are careful while drawing out the recursive tree and summing up the costs, direct proof of a solution to a recurrence.
- Else, you can use it as a guess for substitution method.

RECURSION TREES

Example: $T(n) = 3T(n/4) + \Theta(n^2)$.
Draw out a recursion tree for $T(n) = 3T(n/4) + cn^2$:

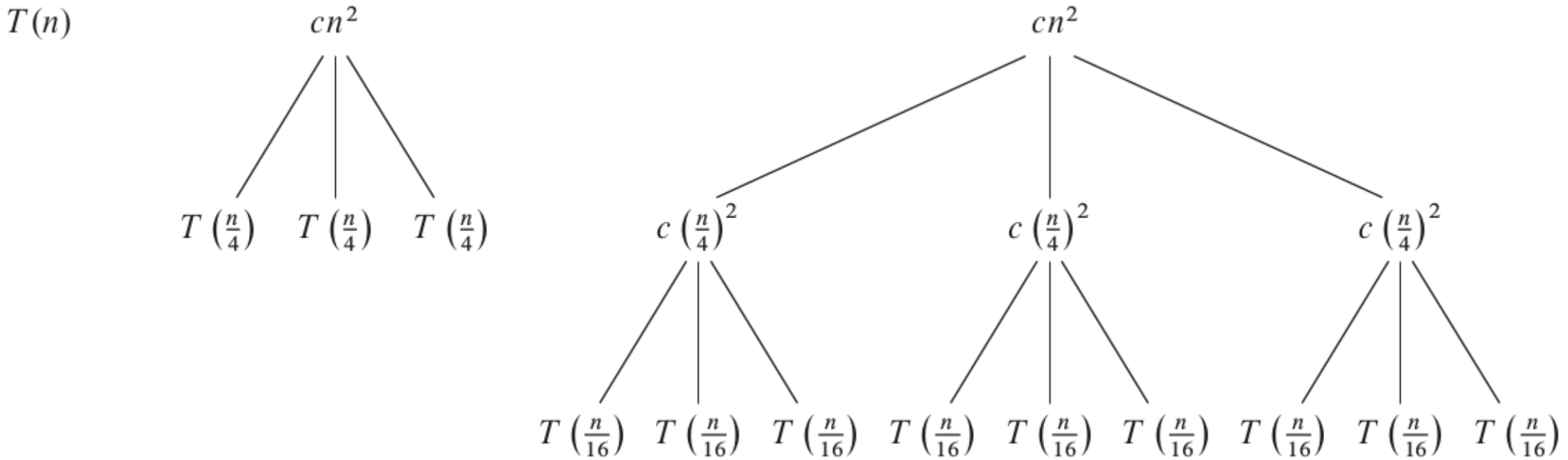
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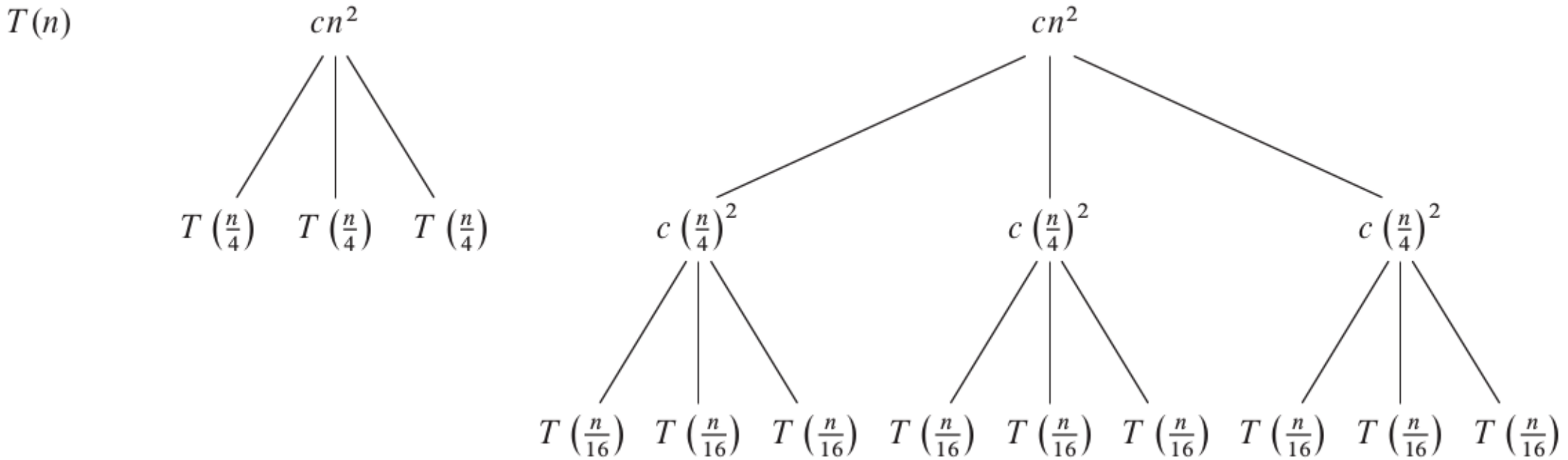
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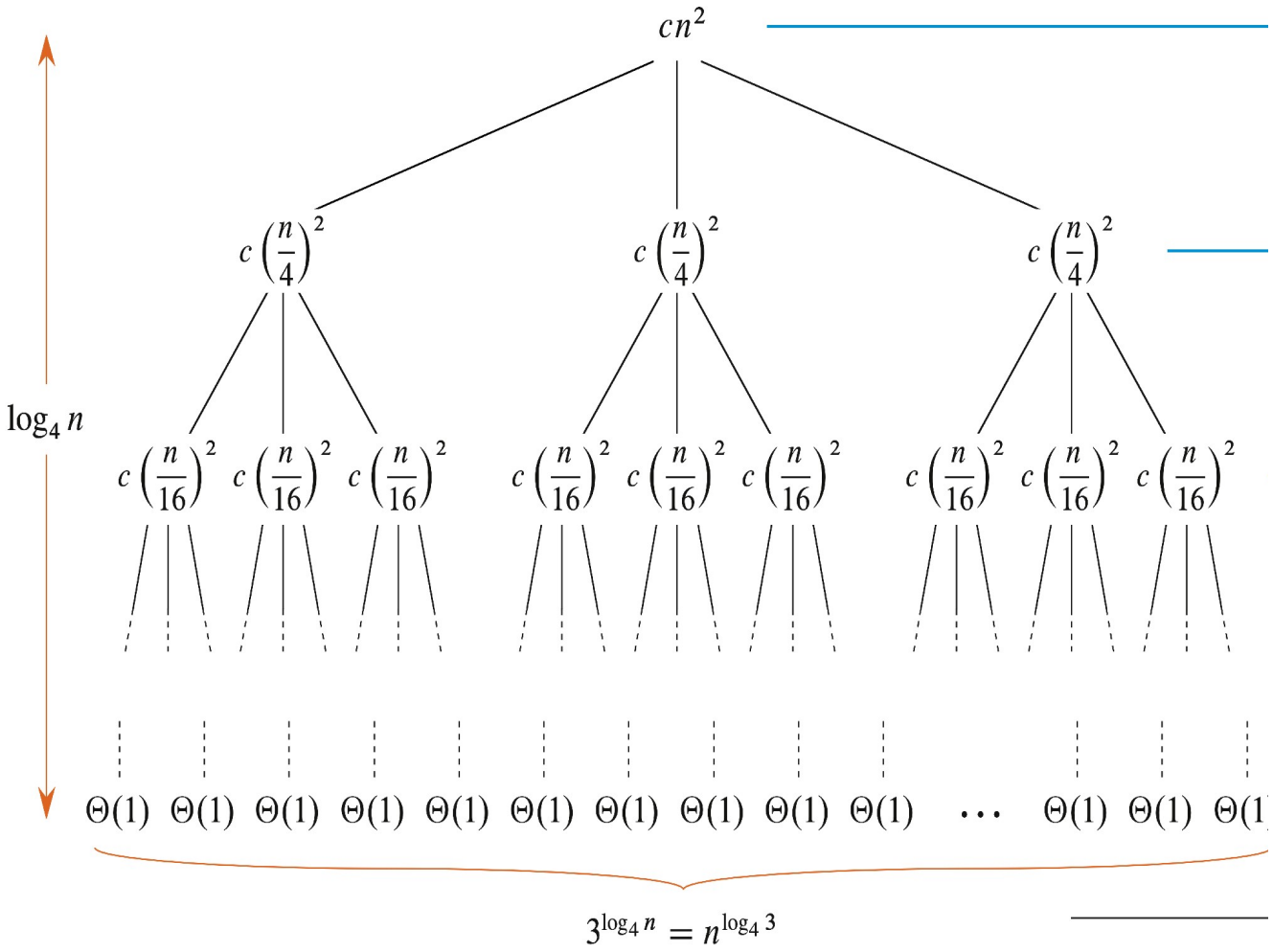


Questions

- Height?
- Number of leaves?

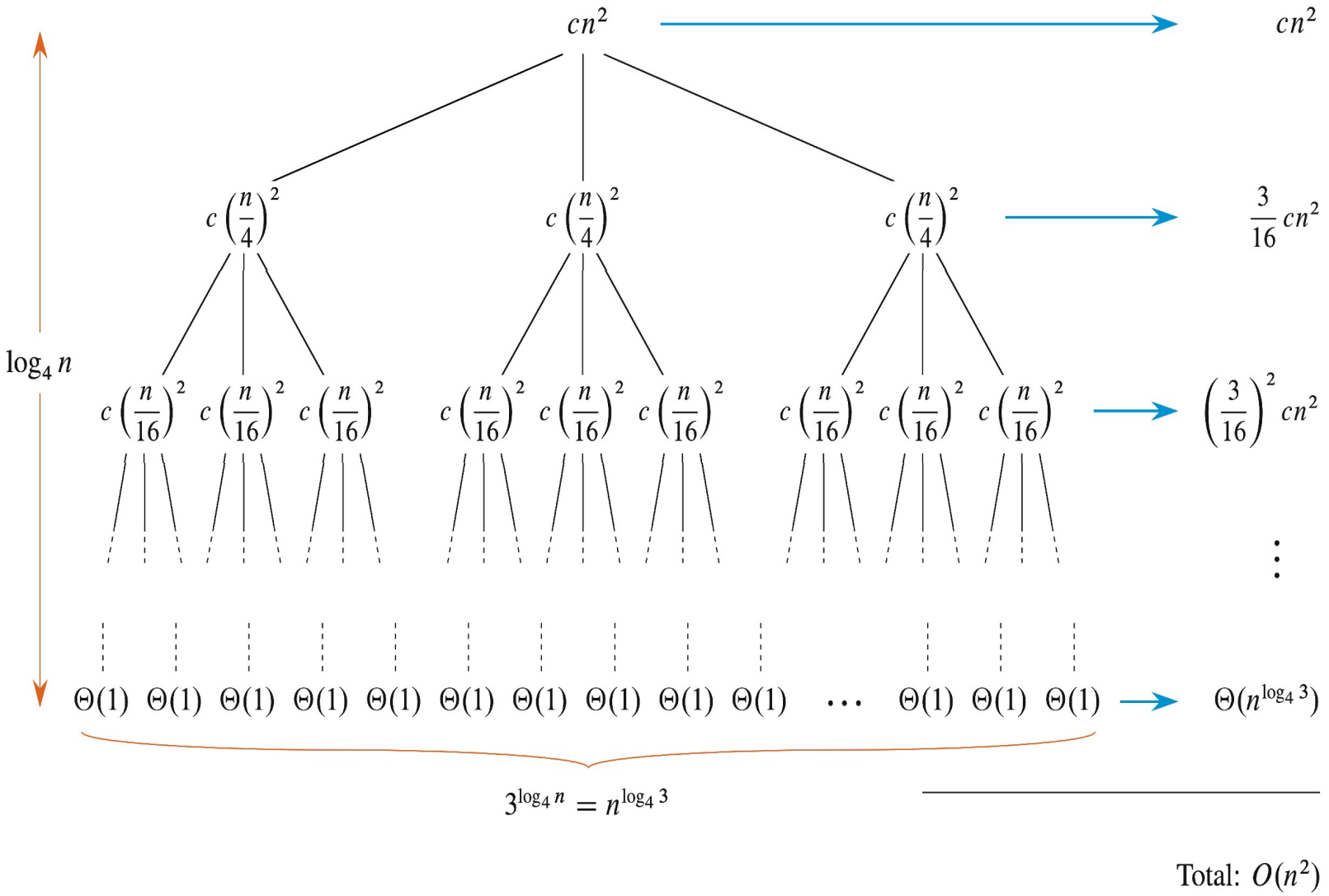


RECURSION TREES



For simplicity, assume that n is a power of 4 and the base case is $\Theta(1)$.

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RECURSION TREES (continued)

Subproblem size for nodes at depth i is $n/4^i$. Get to base case when $n/4^i = 1 \Rightarrow n = 4^i \Rightarrow i = \log_4 n$.

Each level has 3 times as many nodes as the level above, so that depth i has 3^i nodes. Each internal node at depth i has cost $c(n/4^i)^2 \Rightarrow$ total cost at depth i (except for leaves) is $3^i c(n/4^i)^2 = (3/16)^i cn^2$. Bottom level has depth $\log_4 n \Rightarrow$ number of leaves is $3^{\log_4 n} = n^{\log_4 3}$. Since each leaf contributes $\Theta(1)$, total cost of leaves is $\Theta(n^{\log_4 3})$.

Add up costs over all levels to determine cost for the entire tree:

$$
\begin{aligned}
T(n) &= \sum_{i=0}^{\log_4 n - 1} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&< \sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{1}{1-(3/16)} cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{16}{13} cn^2 + \Theta(n^{\log_4 3}) \\
&= O(n^2) \, .
\end{aligned}
$$

Idea: Coefficients of cn^2 form a decreasing geometric series. Bound it by an infinite series, and get a bound of 16/13 on the coefficients.

RECURSION TREES (continued)

Use substitution method to verify $O(n^2)$ upper bound. Show that $T(n) \leq d\,n^2$ for constant $d > 0$:

RECURSION TREES (continued)

Use substitution method to verify $O(n^2)$ upper bound. Show that $T(n) \leq dn^2$ for constant $d > 0$:

$$\begin{aligned} T(n) &\leq 3T(n/4) + cn^2 \\ &\leq 3d(n/4)^2 + cn^2 \\ &= \frac{3}{16} dn^2 + cn^2 \\ &\leq dn^2 , \end{aligned}$$

by choosing $d \geq (16/13)c$. *[Again, we get to name but not choose c, and we get to name and choose d.]*

Can you comment on the lower bound?

RECURSION TREES (continued)

Use substitution method to verify $O(n^2)$ upper bound. Show that $T(n) \le dn^2$ for constant $d > 0$:

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by choosing $d \ge (16/13)c$. *[Again, we get to name but not choose c, and we get to name and choose d.]*

That gives an upper bound of $O(n^2)$. The lower bound of $\Omega(n^2)$ is obvious because the recurrence contains a $\Theta(n^2)$ term. Hence, $T(n) = \Theta(n^2)$.

RECURSION TREES (continued)

Irregular Example:

Inductive hypothesis: $T(n) \leq cn \lg n$ for all $n \geq n_0$. Will choose constants $c, n_0 > 0$ later, once we know their constraints.

$T(n) = T(n/3) + T(2n/3) + \Theta(n)$.

For upper bound, rewrite as $T(n) \leq T(n/3) + T(2n/3) + cn$; for lower bound, as $T(n) \geq T(n/3) + T(2n/3) + cn$.

What does cost at each level specify, in terms of the time taken for the subproblems?

RECURSION TREES (continued)

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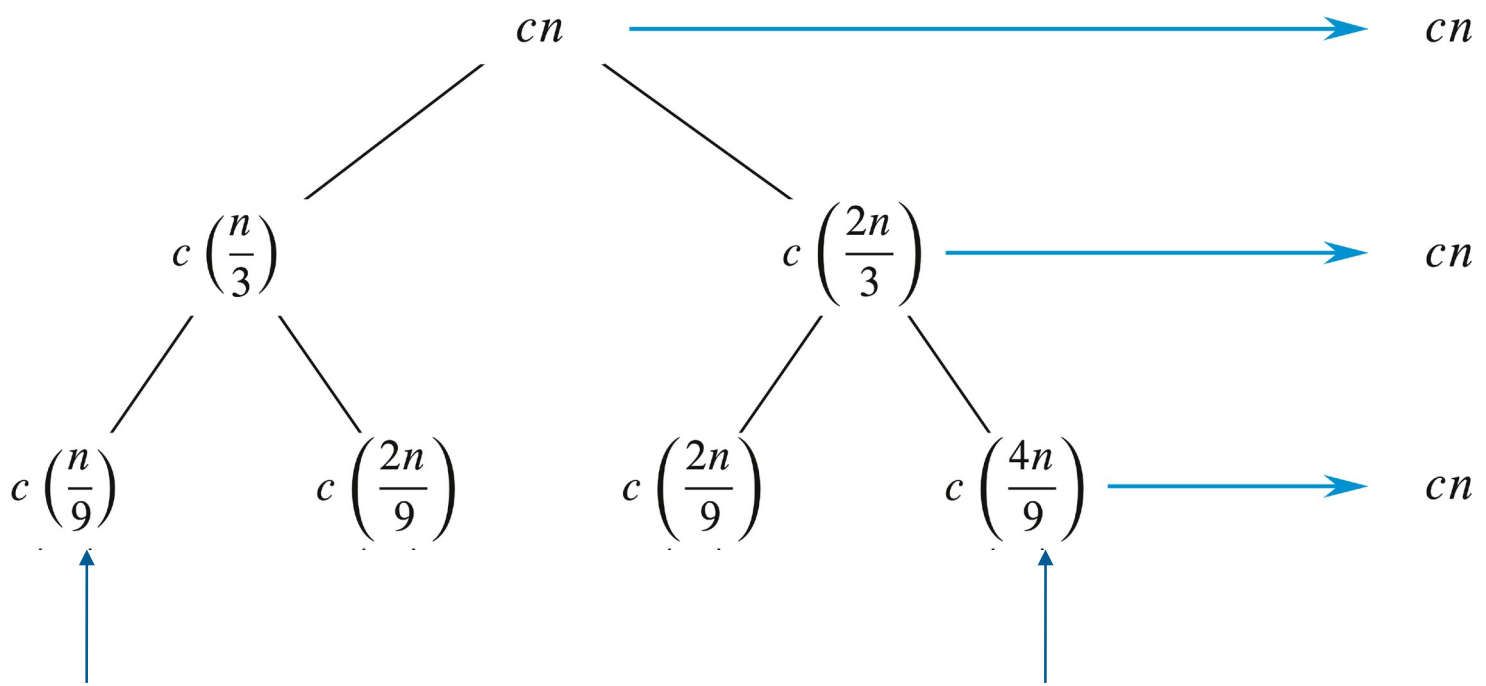
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By summing across each level, the recursion tree shows the cost at each level of recursion (minus the costs of recursive calls, which appear in subtrees):

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

RECURSION TREES (continued)

Irregular Example:

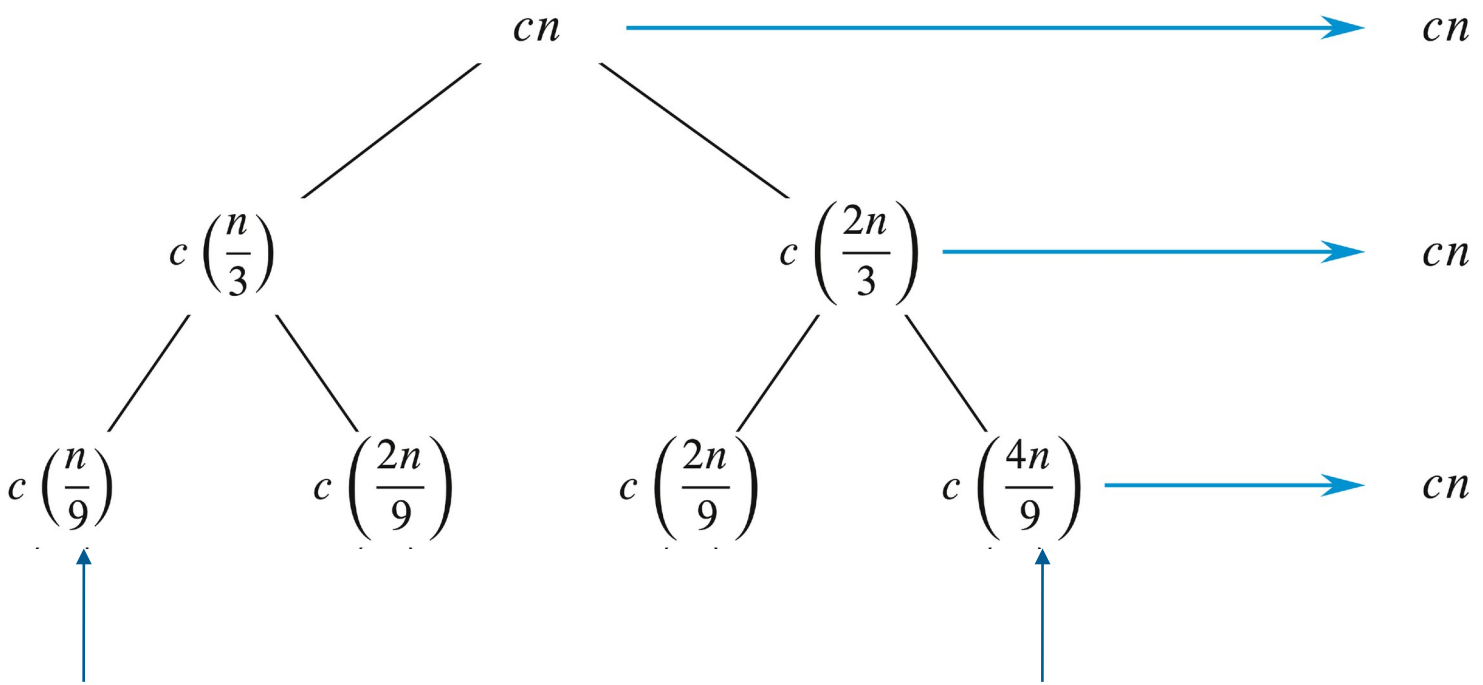


leftmost branch reaches base case after levels

rightmost branch reaches base case after levels

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

RECURSION TREES (continued)



leftmost branch reaches base case after $\log_3 n$ levels

rightmost branch reaches base case after $\log_{3/2} n$ levels

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

RECURSION TREES (continued)

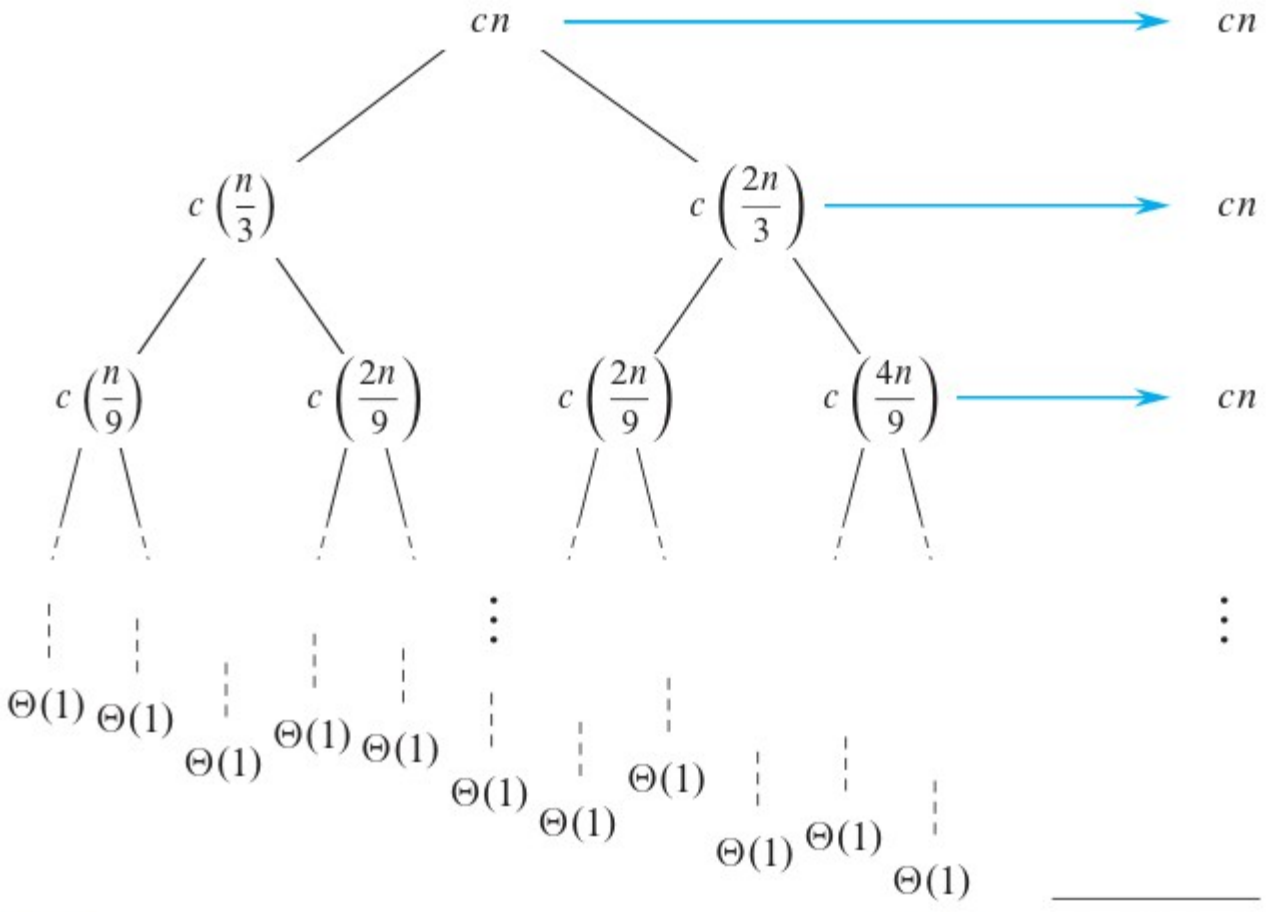
Irregular Example:

Height of the tree?

Cost at each level?

#leaves?

Cost of the entire tree?



$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

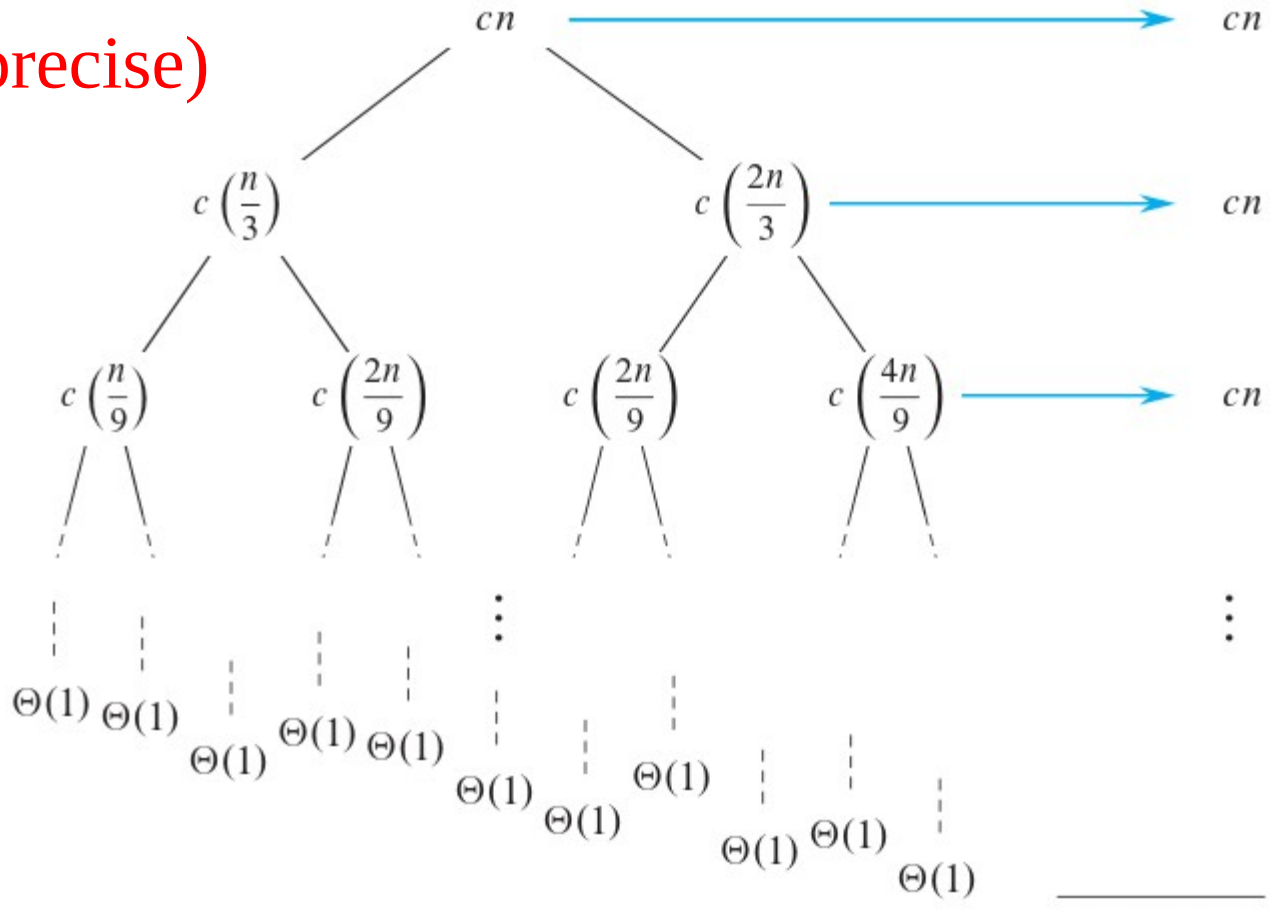
RECURSION TREES (continued)

Height of the tree? **$\log_{3/2} n$** (not precise)

Cost at each level? **cn**

#leaves?

Cost of the entire tree?



Is #leaves 2^h where h= $\log_{3/2} n$?

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

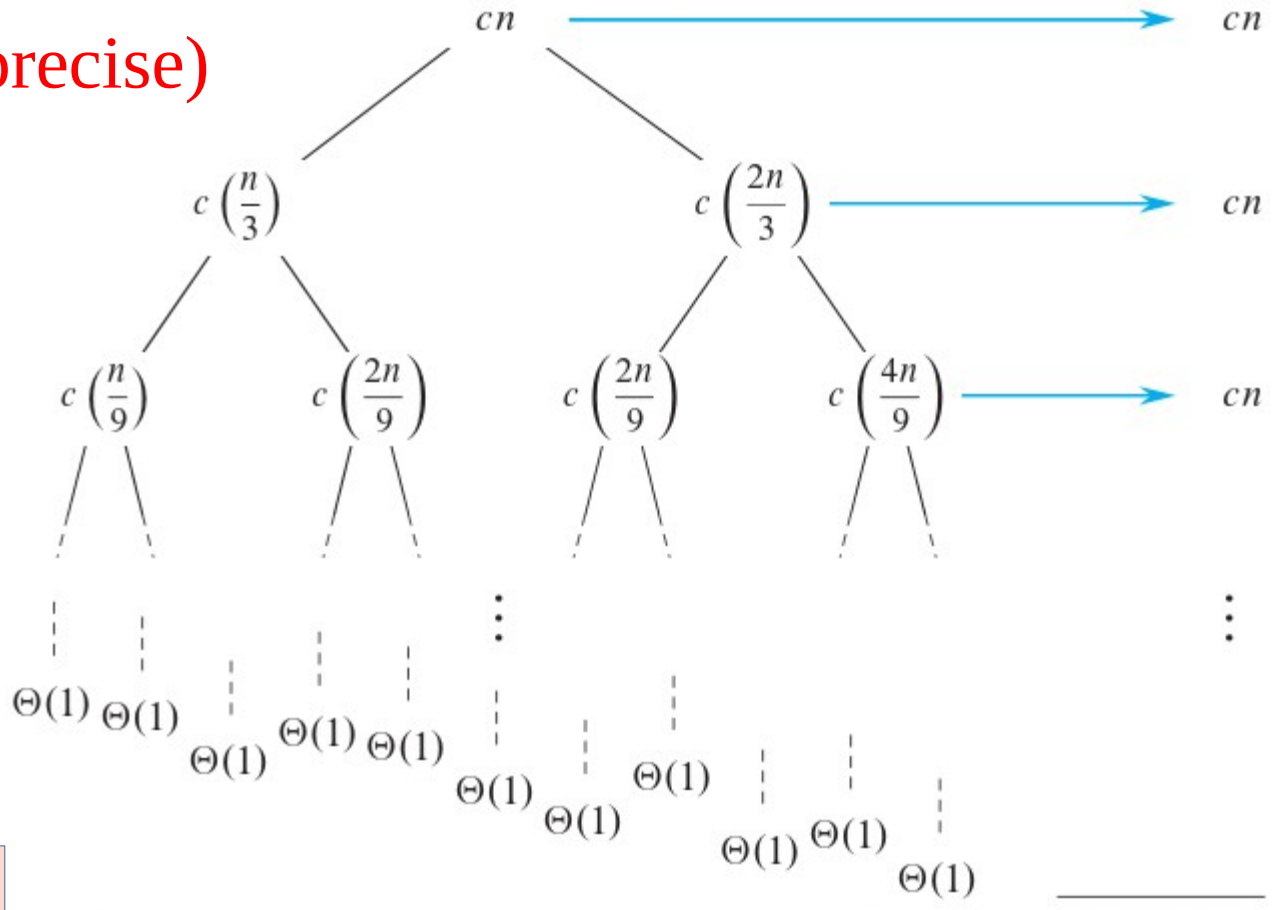
RECURSION TREES (continued)

Height of the tree? **$\log_{3/2} n$** (not precise)

Cost at each level? **cn**

#leaves?

Cost of the entire tree?



Is #leaves 2^h where $h = \log_{3/2} n$?

$\sim 2^{\log_{3/2} n} = n^{\log_{3/2} 2} = \mathbf{n^{1.71}}$

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

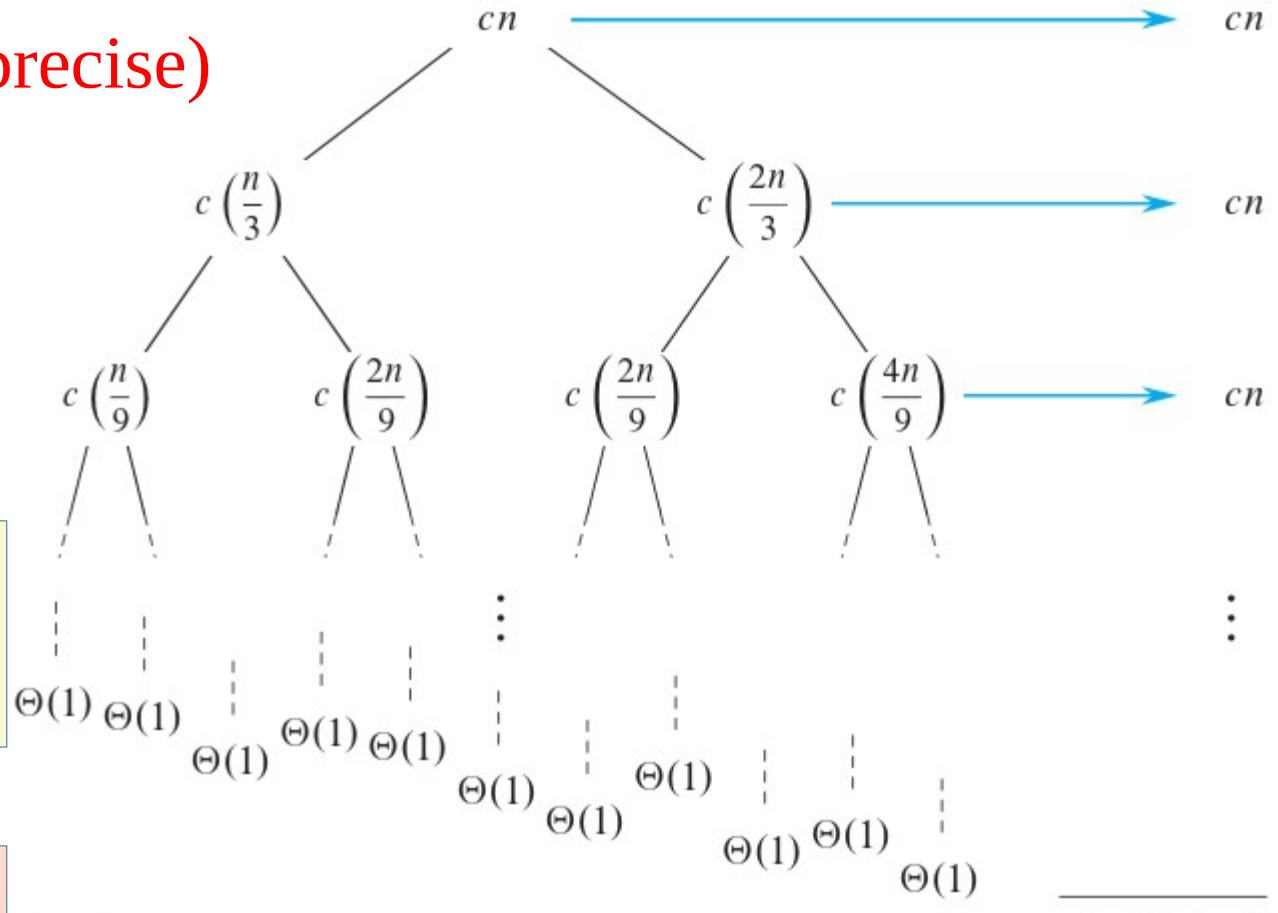
RECURSION TREES (continued)

Height of the tree? **$\log_{3/2} n$** (not precise)

Cost at each level? **cn**

#leaves?

Cost of the entire tree?



Cost at leafs **?** cost at the internal nodes

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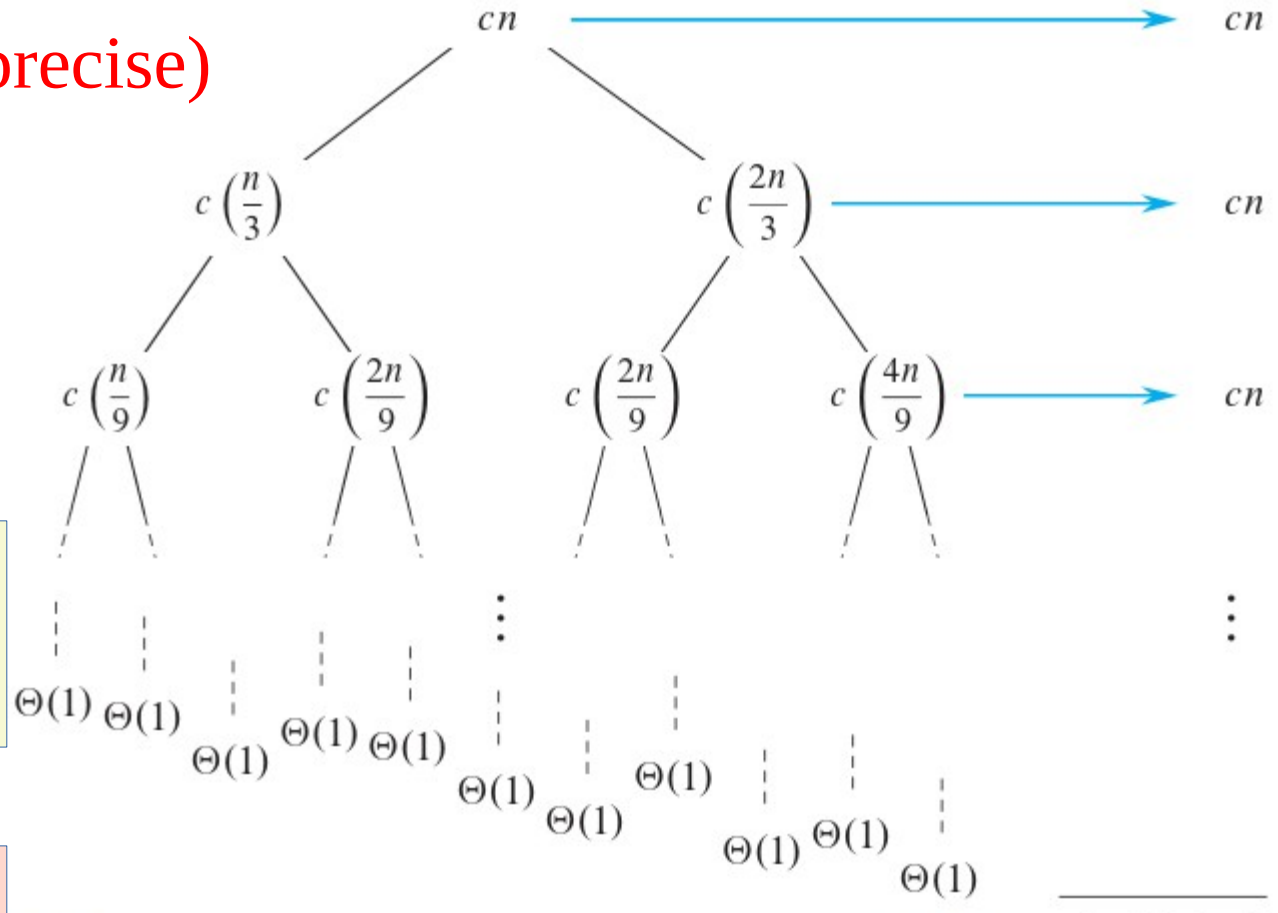
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Cost at each level? **cn**

#leaves?

Cost of the entire tree?



Cost at leafs ? cost at the internal nodes

$n^{1.71}$ $n \log_{3/2} n$

Is #leaves 2^h where $h = \log_{3/2} n$?

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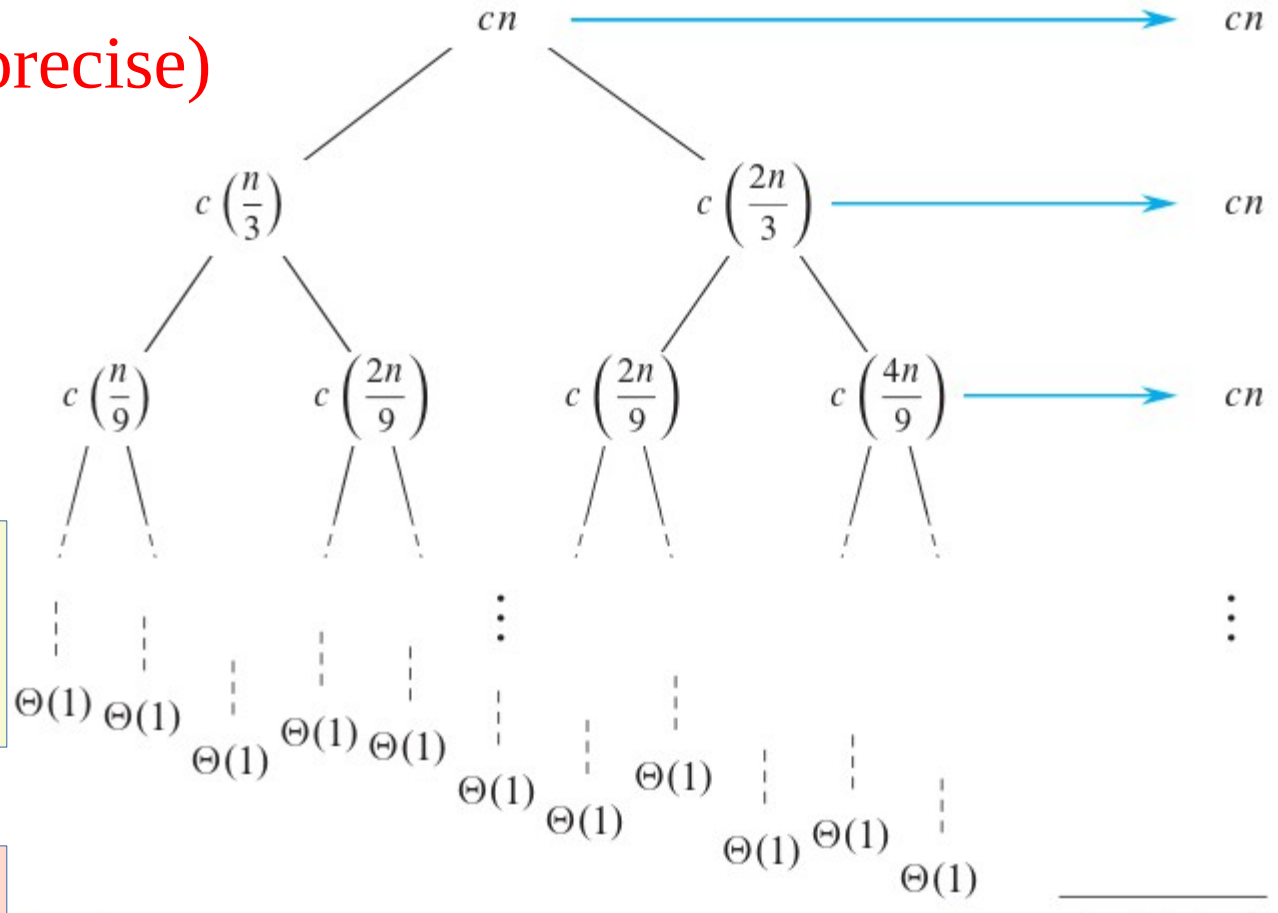
RECURSION TREES (continued)

Height of the tree? **$\log_{3/2} n$** (not precise)

Cost at each level? **cn**

#leaves?

Cost of the entire tree?



Cost at leafs > cost at the internal nodes

$n^{1.71}$ $n \log_{3/2} n$

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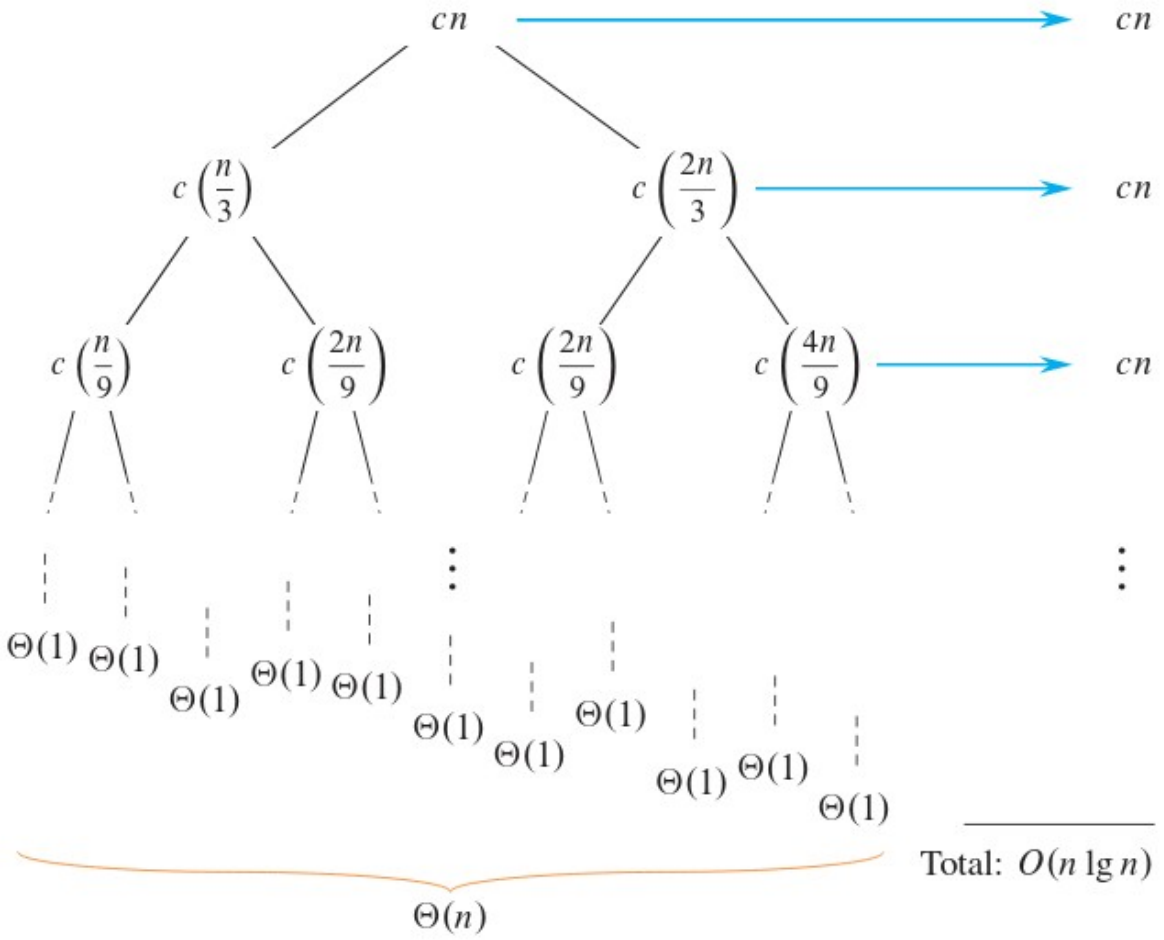
RECURSION TREES (continued)

Height of the tree? **$\log_{3/2}$ n** (not precise)

Cost at each level? **cn**

#leaves? **Assume it is linear**

Cost of the entire tree?



Cost of internal nodes + cost at leaves= $O(n \log_{3/2} n) + O(n)$

$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

RECURSION TREES (continued)

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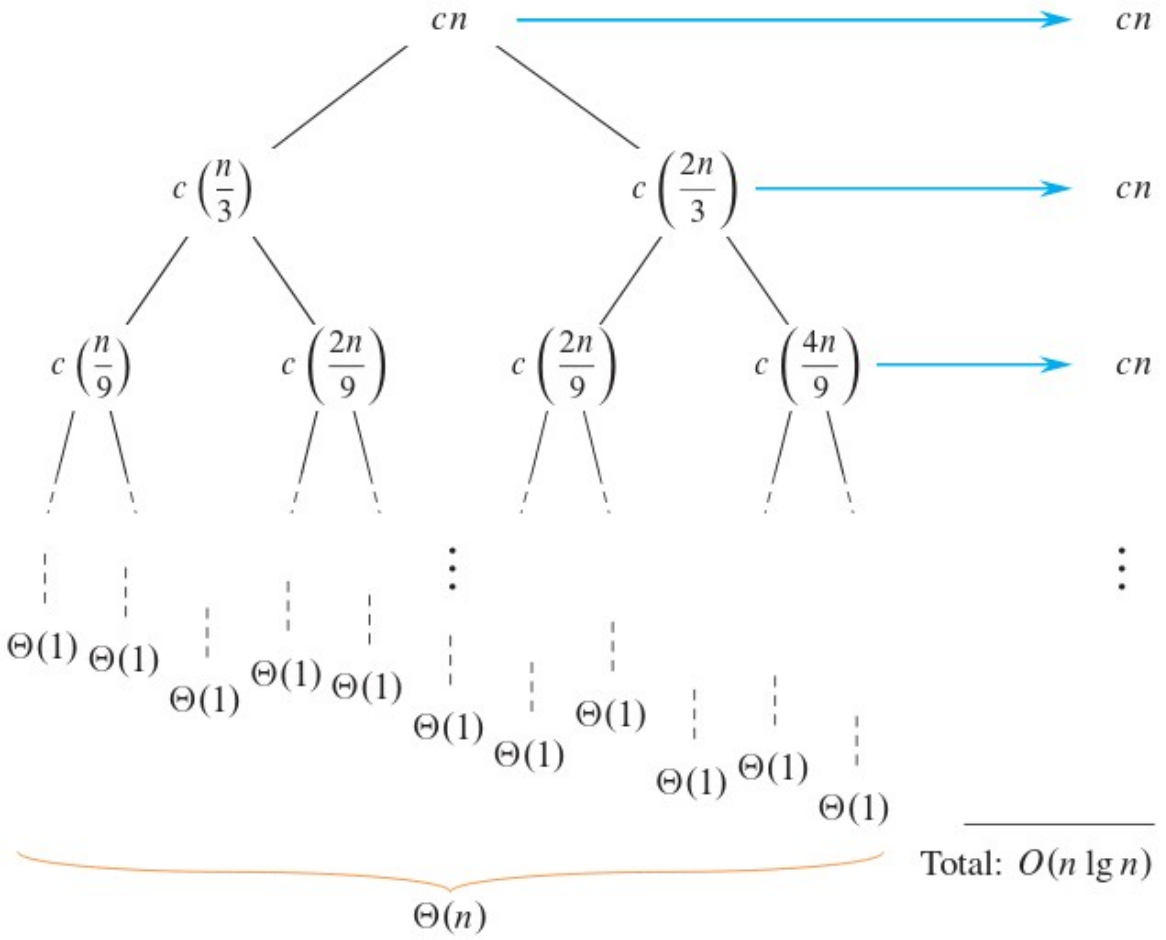
Cost at each level? **cn**

#leaves? **Assume it is linear**

Cost of the entire tree?

D&C approach to find #leaves?

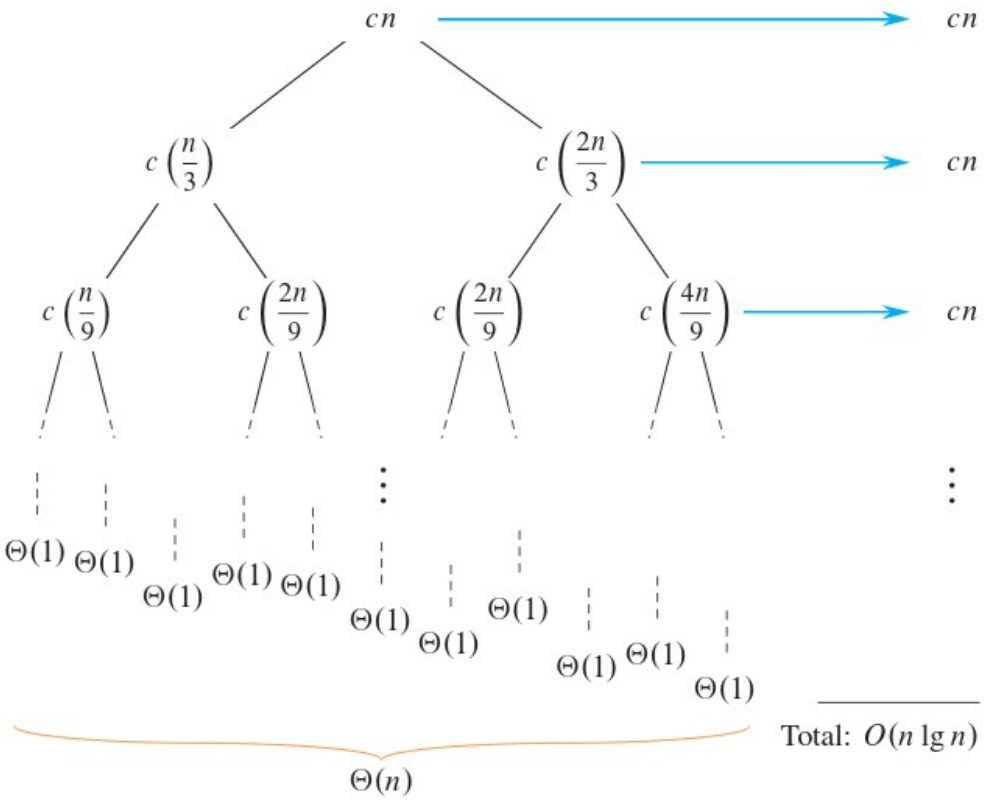
Cost of internal nodes + cost at leaves= $O(n \log_{3/2} n) + O(n)$



$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

#leaves

L(n) =



$$T(n) = T(n/3) + T(2n/3) + \Theta(n).$$

#leaves

L(n) = denotes the number of leaves in the recursion tree of size n.

- What is the base case?
- What is the recurrence?

L(n) = ?

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#leaves

L(n) = denotes the number of leaves in the recursion tree of size n.

- What is the base case? **L(1)=1**
- What is the recurrence?

L(n) = ?

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#leaves

L(n) = denotes the number of leaves in the recursion tree of size n.

- What is the base case? **L(1)=1** (more bases cases depends on the value of n)
- What is the recurrence?

L(n) = L(n/3) + L(2n/3)

RECURSION TREES (continued)

Irregular Example:

- Some levels end up soon, and we can carefully account for the time.
- If we want a good guess to apply **Substitution method**, this is good enough.